HOMER Analysis for Integrating Wind Energy into the Grid in Southern of Algeria

Djohra SAHEB KOUSSA
Centre de Développement des Energies Renouvelables,
CDER,16340,
Algiers, Algeria
d.saheb@cder.dz

Mustapha KOUSSA
Centre de Développement des Energies Renouvelables,
CDER,16340,
Algiers, Algeria
m.koussa@cder.dz

Abstract*—Wind is an alternative energy resource, which is clean, inexhaustible and environmentally friendly. In this paper, an investigation is made on large-scale operations of 95 MW per day wind energy system (WES) as a grid power generation consisting of wind energy. A comparison is drawn between a grid-connected wind energy system GCWES and a standard grid operation focusing on environmental and economic impacts. Emissions and the wind energy generation fraction (RF) of total energy consumption are calculated as the main environmental indicator. Costs including net present cost (NPC), cost of energy (COE) and payback period are calculated for economic evaluation. To simulate the WES, the hourly wind speed data from InSalah (27° 09' 00" N, 2° 27' 38 E) of Algeria, are used as an example of a typical arid climate. HOMER is used for simulation. It is found that the optimum results of GCWES show a 19% reduction of emissions including CO2, SO2 and NOx. The RF of the optimized system is 19%. It is also found that the reduced NPC and COE are only equal to about 81% of energy consumption from standard grid and a payback is equal to 5.77 years. In addition, through a set of sensitivity analysis it is found that the wind speed has more effects on the environmental and economic performance of a GCWES.

Keywords-wind; Grid; COE; Emissions; RF; HOMER;

I. INTRODUCTION

Since the demand for energy and specifically electricity, has increased so dramatically over the last 100 years. It has now become important to consider the environmental impacts of energy production. In the past, high standards of living and "modern lifestyles" were based on increased energy consumption. Today, statistics from highly developed countries show that standards of living can increase independently of energy consumption if energy efficiency measures are introduced [1].

In 1999, the world global demand for electricity was about $14{,}764 10^9$ kWh. This demand was met mainly through fossil fuels and nuclear power. Renewable energies, besides hydropower with 18%, only had a 2% share [1].

Therefore, the Kyoto protocol is the first attempt to limit carbon dioxide emissions through an international convention. Wind energy and other renewable energy sources provide a way to meet increasing energy demand without compromising the environment and contributing to climatic disaster [1].

In this context, in the last decade a lot of studies related to the electricity production via renewable energy have been developed in many countries worldwide by researchers [2 - 6].

Ismail et al [2] have proposed a techno-economic analysis and the design of a complete hybrid system, consisting of photovoltaic (PV) panels, a battery system and a diesel generator as a backup power source for a typical Malaysian village household is presented. The specifications of the different components constructing the hybrid system were also determined. A scenario depending on a standalone PV and other scenario depending on a diesel generator only were also analyzed. A simulation program was developed to simulate the operation of these different scenarios.

Engin, M [3] has developed a sizing procedure for hybrid system with the aid of mathematical models for photovoltaic cell, wind turbine, and battery that are readily present in the literature. This sizing procedure can simulate the annual performance of different kinds of photovoltaic-wind hybrid power system structures for an identified set of renewable resources, which fulfills technical limitations with the lowest energy cost. The output of the program will display the performance of the system during the year, the total cost of the system, and the best size for the PV-generator, wind generator, and battery capacity. Security lightning application is selected, whereas system performance data and environmental operating conditions are measured and stored.

Essalaimeh et al [4] have showed an experimental investigation of using a combination of solar and wind energies as hybrid system for electrical generation under the Jordanian climate conditions. The generated electricity has been utilized for different types of applications and mainly for space heating and cooling. The system has also integration with grid connection to have more system that is reliable. Measurements included the solar radiation intensity, the ambient temperature, the wind speed and the output power from the solar PV panels and wind turbine.

Fay and Udovyc [5] have summarized the findings of an informal survey conducted on the most important characteristics of a successful wind-diesel hybrid power project in small remote rural communities. The survey was done to help guide

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socioeconomic research in Alaska on community capacity to ensure sustainable projects.

In another study, Celik has [6] presented an optimization and techno-economic analysis of autonomous photovoltaic – wind hybrid energy systems in comparison to single and wind systems.

In Algeria, a number of studies on hybrid renewable energy systems (HRES) was accomplished [7-13].

Khelif et al [7] have undertaken a feasibility of a hybrid PV/Battery/Diesel power plant using real meteorological data equipment costs to show the possibility of modifying a stand-alone Diesel generator installation located in AFRA, south of Algeria, into a hybrid system.

D. Saheb-koussa and al [8] have elaborated a technico-economic study of the hybrid system consisting of wind and photovoltaic with battery storage, in which the diesel generator is added to ensure continuous power supply and to take care of the intermittent nature of wind and photovoltaic. This serves as an additional tool that helps choosing the best system (wind or photovoltaic installation) when the two possibilities are technically possible. The choice was based on the determination of the option, which corresponds to the least cost and to the best performances.

Kaabache et al [9] have proposed an integrated PV/wind hybrid system model, which utilizes the iterative optimization technique following the Deficiency of Power Supply Probability (DPSP), the Relative Excess Power Generated (REPG), the Total Net Present Cost (TNPC), the Total Annualized Cost (TAC) and Break-Even Distance Analysis (BEDA) for power reliability and system costs. The flow chart of the hybrid optimal sizing model is also illustrated with this merged model.

In [10] a power management control (PMC) for a hybrid system that comprises wind and photovoltaic generation subsystems, a battery bank, and an AC load has been developed.

Himri et al [11] have presented techno-economic assessment for off-grid hybrid generation systems of a site in south western Algeria. The HOMER model are used to evaluate the energy production, life-cycle costs and greenhouse gas emissions reduction.

In [12] Diaf et al have presented a methodology to perform the optimal sizing of an autonomous hybrid PV/wind system. The methodology aims at finding the configuration, among a set of systems components, which meets the desired system reliability requirements, with the lowest value of levelized cost of energy. Modelling a hybrid PV/wind system was considered as the first step in the optimal sizing procedure.

Nevertheless, this paper analyzed the environmental and economic benefits of GCWES used in the arid environment. In Salah is taken as an example of the famous Algerian the arid land. The climatic data is used to simulate the long-term implementation of the system.

II. The Grid Connected Wind Energy System Configuration

The studied energy system consists of wind energy components working together with a non-renewable energy component whereby in this case, the non-renewable energy component is the grid. Figure 1 shows the configuration of grid-connected wind system in HOMER.

The wind energy component will generate electricity to be used by the load. However, when the wind energy supply is insufficient, electricity will be bought from the grid. When the wind energy generates excess electricity, it will be sold back to the grid using Feed in Tariff (FiT) rates.

All the data involved to analyze the potential of the studied system based on a case study InSalah are discussed in the present section. The energy components used in the configuration presented below are also discussed in the following subsections.

A. Load Demand

In this study, a simple story of a small village connected to the classical energy distribution network is considered. Nonetheless, it is equipped with a number of devices to the comfort of its inhabitants like hospitals, plant and schools. Additionally, it is assumed to be occupied all over the year and that its devices require the standard rated voltage (220V AC, 50 Hz).

Figure 2. shows samples schematic load curve of the considered community. it represents, for each hour of the day, how much power is needed in kilowatt. Typically, in a small village, there one peak loads of about 4.6MW, when people turn on their lights in the evening.The annual energy consumption of the village is 24496.8 MWh with an average daily energy consumption of 67MWh.

B. Wind energy resources

The wind Hourly data has been used. These data was obtained from METEONORM 7 this last one are illustrated on Figure 3(a). Therefore, as shown in Figure 3(b) the average wind speed ranges is from 4.689 m/s to 6.331m/s with an annual average of 5.62 m/s. From the same figure, it can be seen that the wind speed is highest during the months of March and April. Therefore, in the other months the wind speed is not low which favorites to consider the selected site for the wind energy production.

C. Grid connected wind energy system

The GCWES consists of grid, and wind turbines. For the economic analysis, HOMER includes the initial cost, replacement cost, and operating and maintenance cost.

- Wind energy conversion system

Availability of electrical energy from the wind turbine depends significantly on wind speed. Vestas V82 wind turbine

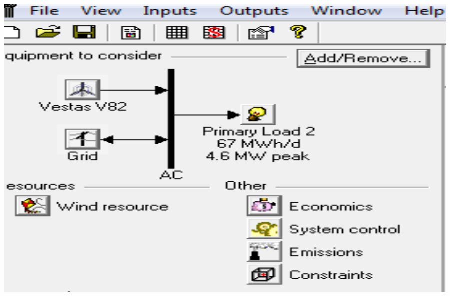

Figure 1. Configuration of GCWES in HOMER

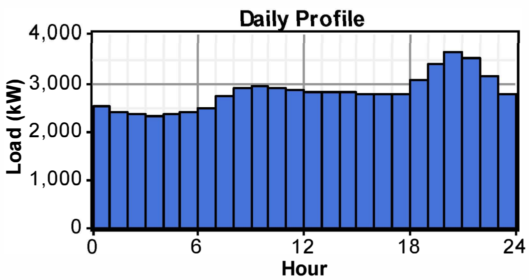

Figure 2. Hourly load profile

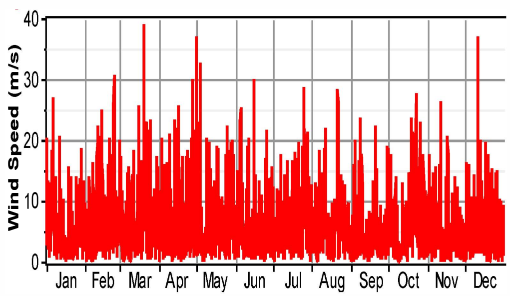

Figure 3(a). Hourly wind speed

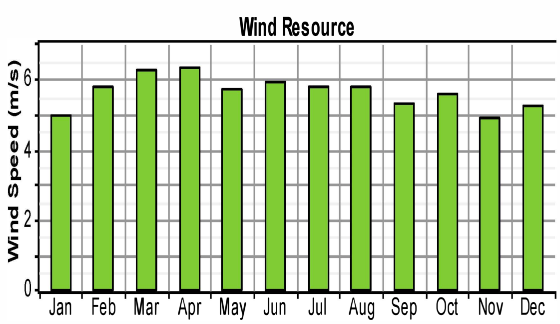

Figure 3(b). Monthly averaged wind speed data

model is considered (Figure 4). It has a rated capacity of 1.65kW and provides 690V AC as output with 50Hz frequency.

Cost of one unit is considered to be 1262190 $ while replacement and maintenance costs are taken as 1262190 $ and 2000$. To allow the simulation program find an optimum solution, provision for using 0 (without turbine), 1 or 2 units is given. Lifetime of a turbine is taken to be 20 years.

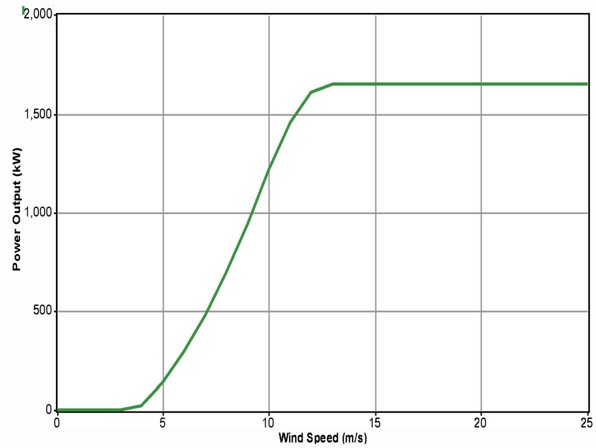

Figure 4. Wind turbine Vestas V80 versus wind speed

- Grid

In the present case, the grid is a principal source by which it acts with the wind energy system to satisfy the load demand. Therefore, there is grid power price, which is the electricity, bought from the grid and the price of electricity sold to the grid, also known as feed in tariff (FIT). The electricity tariff for the wind system in Algeria is 300% of base rate.

Table 1 illustrated the samples of FIT application in Algeria[13]. It can be remembered that Algeria is depending on base rate and technology.

TABLE 1. SAMPLES OF FIT APPLICATION IN ALGERIA

	% of Base Rate	$c/kWh
Base Rate		4.82
Wind Energy	300%	14.5
Solar Photovoltaic	300%	14.5
Concentrating Solar	200%	9.57
Hydro	100%	4.82
Waste	200%	9.57

III. ASSESSMENT CRITERIA

The analysis on the GCWES in terms of economic feasibility is based on net present cost (NPC) and payback period. HOMER assesses the technical viability of the wind energy system by determining when the system participate to serve the load adequately. Then, it estimates the NPC of the system and classes all the system according to total net present cost NPC[13].

A. Net present Cost (NPC)

The total NPC assesses all costs that meet with the lifetime. The costs include initial costs, replacement cost, operating and maintenance costs and cost of electricity from the grid. Total NPC is calculated by using the following equation:

$$NPC = \frac{C_{ann,tot}}{CRF(i, R_{proj})} \quad (1)$$

Where Cann,tot is the total annualized cost [$/yr];

CRF( ) is the capital recovery factor;

i is the interest rate [%];

Rproj is the project lifetime [yrs].

The nominal interest rate and annual inflation rate in Algeria as of 2014 are 5.9% and 1.8 respectively [13].Therefore, the annual real interest rate is 2.25%.

B. Payback period

The payback period is an indication of how long it would take to recover the difference in investments costs between the current system and the base case system. In this case, the current system would be the GCWES and the base case system would be the grid-only system.

IV. RESULTS AND ANALYSIS

A. Optimisation results

The categorized result for In Salah (Table 2) in which the two systems(the grid only and the wind-grid system) architecture types are found is listed in increasing order of total Net Present Cost(NPC). There are two different feasible system architecture starting from wind –Grid system to the grid only. From Table 2 it is indicate that wind-grid system with 1 unit of 1.65 kW wind turbine is the most viable system to be applied as it has the lowest NPC. Figure 5 shows the electricity production from the wind grid system.

TABLE 2.CATEGORIZED OPTIMIZATION RESULTS

Sensitivity Results | Optimization Results

Sensitivity variables

Wind Speed (m/s) 5.62 Rate 1 Power Price ($/kWh) 0.048 Rate 1 Sellback Rate ($/kWh) 0.146

Double click on a system below for simulation results. Categorized Overall Export... Details...

V82	Grid (kW)	Initial Capital	Operating Cost ($/yr)	Total NPC	COE ($/kWh)	Ren. Frac.
1	5000...	$ 1,262,190	962,804	$ 13,570,056	0.043	0.19
	5000...	$ 0	1,17 ,894	$ 15,031,877	0.048	0.00

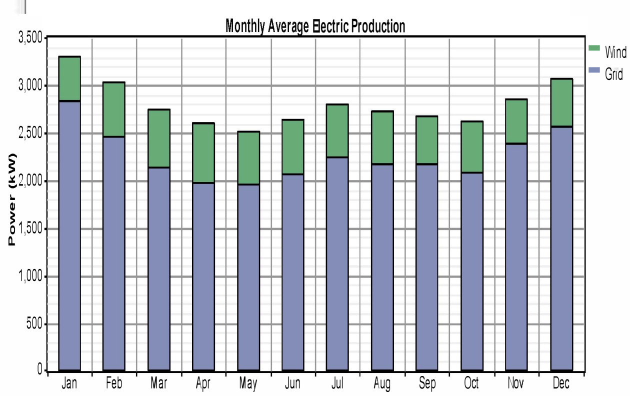

Figure 5. Average Monthly Electric Production from each source

B. Grid –only system

The NPC of a grid only system is 15,031,877 $ at a cost of electricity of 0.048$/kWh. The total electricity produced by the grid is 24,497 MWh/yr.

C. Wind –Grid system

At FIT rate equivalent to Wind-grid system of 0.145$/kWh, the Wind –grid system which consists of one unit of 1.65 kW wind turbine is the most viable system to be implemented. The NPC is 13,570,056$ which is approximately 15% lower than the NPC of Grid only system. Figure5 shows that the electricity produced the wind turbine is 19% and grid 81%. This show that the renewable fraction of the system is small compared to the conventional grid which means the avoid of problems that can result by the connection of wind turbines to the electrical grid.

The Figure 6 shows the power production of system component with primary electrical load 67MWh/d during the January- February and June-July respectively. Electrical energy produced by the grid decreases when the wind energy increases. In the case of no wind, the electric grid only will ensure energy production the supplying of the load. Therefore, the GTWES is feasible from economic point of view. In addition, payback period for present system is obtained as 5.57 years (Table 3) which ensures approximate 6 years of net income for a project life of 25 years.

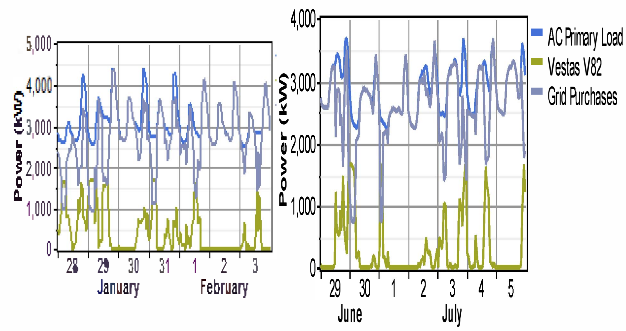

Figure 6. Power production of system components during the January- February and June-July with primary electrical load 67MWh/d

TABLE 3. ECONOMIC COMPARISON BETWEEN GCWES AND GRID ONLY SYSTEM

	V82	Grid (MW)	Initial capital	Total NPC
Base Case		5000	$0	$15,031,877
Current System	1	5000	$1,262,190	$13,570,056

Metric	Value
Present worth	$1,461,823
Annual Worth	$114,354/yr
Return on investment	17.0%
Internal rate of return	17.1%
Simple Payback	5.57yrs
Discounted payback	6.98 yrs

The amount of yearly emissions of the GCWES and the grid only system are given in Table 7. In this case, amount of emission of CO2, SO2 and nitrogen oxides reduced to 12,472,408kg/year, 54,073 kg/year, and 26,445kg/year respectively. These values are very much lower than the grid-only system. Hence, this system is beneficial in the context of clear environment, too.

TABLE 4.YEARLY EMISSIONS OF GCWES AND GRID ONLY SYSTEM

Pollutant	Emissions of the GCWES (kg/yr)	Emissions of the grid only system (kg/yr)
Carbon dioxide	12, 472,408	15,482,610
Carbon monoxide	0	0
Unburned hydrocarbons	0	0
Particulate matter	0	0
Sulfur dioxide	54,073	67,124
Nitrogen oxides	26,445	32,827

V. SENSITIVITY ANALYSIS

The sensitivity analysed was carried out of wind speed and grid power price in order to incorporate possible variations of these variables in the future as shown in Figure 7. GCWES is still cost effective and no longer sensitive to grid power price. If the wind speed is equal or superior to 4.75m/s GCWES still economically feasible.

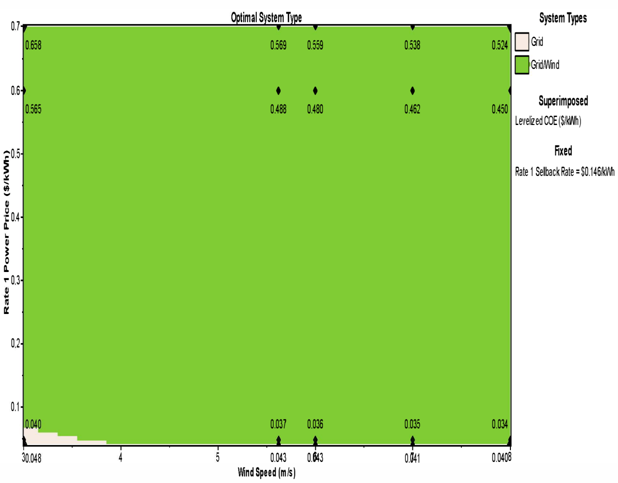

Figure 7. Result of sensitivity analysis

VI. CONCLUSION

In this paper, an economic and environmental analysis of two different configurations that are the grid only system and the wind-grid system has been presented. A large-scale load data and hourly wind speed data during one year were used for the assessment, which was conducted using HOMER software tools. The assessment criteria chosen were net present cost and payback period. The simulation showed that Wind –grid system is the most viable system economically. At fixed FIT rate of 0.145$, the NPC of grid connected one unit of 1.65 kW wind turbine is 13,570,056$ with the RF of the optimized system (GCWES) of 19%. However, for the same FIT rate cited previously, the payback period for GCWES is 5.77 years. From the environmental standpoint, it is found that the optimum results of GCWES show a 19% reduction of emissions including CO2, SO2 and NOx. In addition, through a set of sensitivity analysis it is found that the wind speed has more effects on the environmental and economic performance of a GCWES.

REFERENCES

[1] R.Gash, J.Twele." Wind Power Plants" Fundamentals, Design , Construction and operation. Solarpraxis AG, Germany and James and James London. 2002

[2] Ismail, M. S ; Moghavvemi, M. ; Mahlia, T. M. I..Techno-economic analysis of an optimized photovoltaic and diesel generator hybrid power system for remote houses in a tropical climate. Energy conversion and management 2013 Volume: 69 Pages: 163-173

[3] Engin, M .Sizing and Simulation of PV-Wind Hybrid Power System. International journal of photoenergy, Volume 2013 (2013), 10 pages http://dx.doi.org/10.1155/2013/217526

[4] Essalaimeh, S., Al-Salaymeh, A., Abdullat, Y. Electrical production for domestic and industrial applications using hybrid PV-Wind system: Energy conversion and management 2013, 65 : 736-743.

[5] Fay G.,Udovyk N. Factors influencing success of wind-diesel hybrid systems in remote Alaska communities: Results of an informal survey. Renewable Energy, 2013, 57, Pages 554–557.

[6] Celik A.N. Optimisation and techno-economic analysis of autonomous photovoltaic–wind hybrid energy systems in comparison to single photovoltaic and wind systems. Energy conversion and Management, 2002, 43(18), pp. 2453–2468

[7] Khelif, A, Talha, A ,Belhamel, M , Arab, A.H . Feasibility study of hybrid Diesel-PV power plants in the southern of Algeria: Case study on AFRA power plant. International Journal of Electrical Power & Energy Systems 2012, 43 (1), 546-553.

[8] D.Saheb-Koussa, M.Haddadi et M.Belhamel, Economic and technical study of a hybrid system (wind–photovoltaic–diesel) for rural electrification in Algeria, Appl. Energy, 86 (2009) 1024-1030.

[9] Kaabache A., M.Belhamel et R.Ibtiouen. Techno-economic valuation and optimization of integrated photovoltaic/wind energy conversion system. Solar Energy, 2011 85(10), 2407– 2412

[10] Rekioua, D et Rekioua, T . Modeling, simulation and power management of a stand-alone wind Photovoltaic energy system Computational intelligence in busness and economics World Scientific Proceedings Series on Computer Engineering and Information Science, 2010, 3, 589-596.

[11] Himri, Y., Boudghene Stambouli ,A., Draoui, B., Himri, S. Techno-economical study of hybrid power system for a remote village in Algeria. Energy 33, July 2008, Pages 1128–1136

[12] Diaf, S., Notton, G., Belhamel, M., Haddadi, M., Louche, A. Design and techno – economical optimization for hybrid PV/wind system under various meteorological conditions. «Applied Energy, 85.10 (2008): 968-987.

[13] Mukhtaruddin, R. N. S. R.; Hassan, M. Y.; Rahman, H. A. Economic analysis of grid-connected hybrid photovoltaic-wind system in Malaysia 2013 / IEEE, pp._577 – 583.